Open-Ended Quadrupedal Locomotion through UED

An empirical study of UED algorithms in MuJoCo Playground

Group 6

The Locomotion Challenge

Dense Reward Shaping

Dense reward shaping enforces stability on mathematically perfect, flat topographies. However, it forces agents to optimize for specific numerical artifacts, making them highly rigid.

The gaits learnt may not be robust to varying terrains and input perturbations like low friction, unmodeled payloads, and actuator noise.

The Generalization Gap

Traditional manually-designed curricula rely on human-tuned schedules to introduce obstacles. This creates an artificial "task ceiling."
To cross highly unstructured terrain without explicit shaping, we need open-ended algorithms that can autonomously expand the training frontier and drive emergent robustness.

Proposed Approach

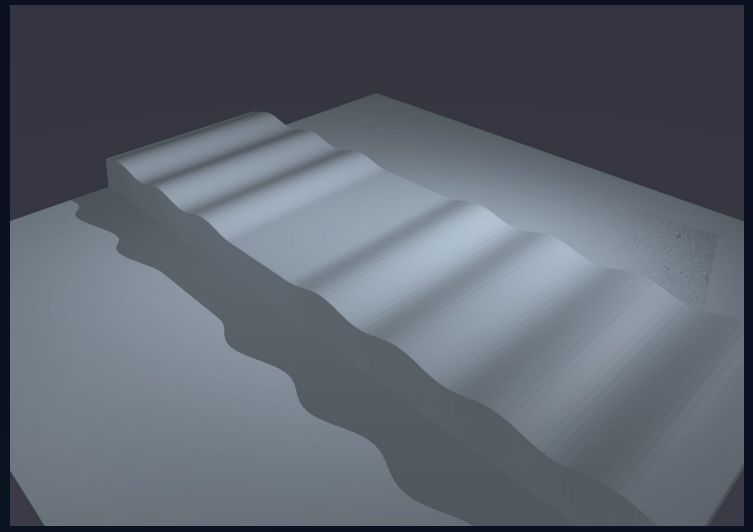
Applying Unsupervised Environment Design (UED) to continuous-control legged locomotion on the Unitree Go1 quadruped.

- **Regret-Driven Discovery:** Leveraging agent performance to dictate terrain complexity.
- **Minimal Reward Shaping:** Stripping away kinematic penalties in favor of sparse forward-velocity rewards.
- **Hardware-Accelerated Env:** Utilizing MuJoCo Playground for rapid, massively parallel simulation.
- **Algorithms Evaluated:**
  - Prioritized Level Replay (PLR) and ACCEL compared against Domain Randomization (DR)
  - UED policies compared to Dense reward shaping based policies



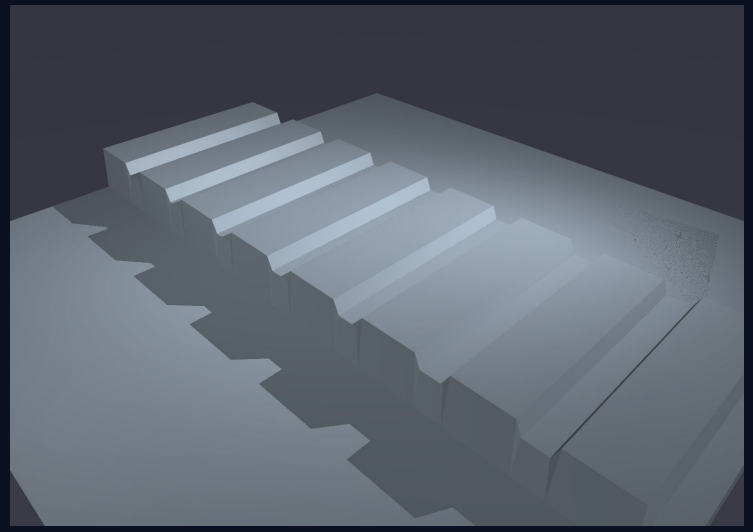
Procedural Track Generation

Evaluating spatial generalization across procedurally generated topological features controlled by a continuous parameter vector.



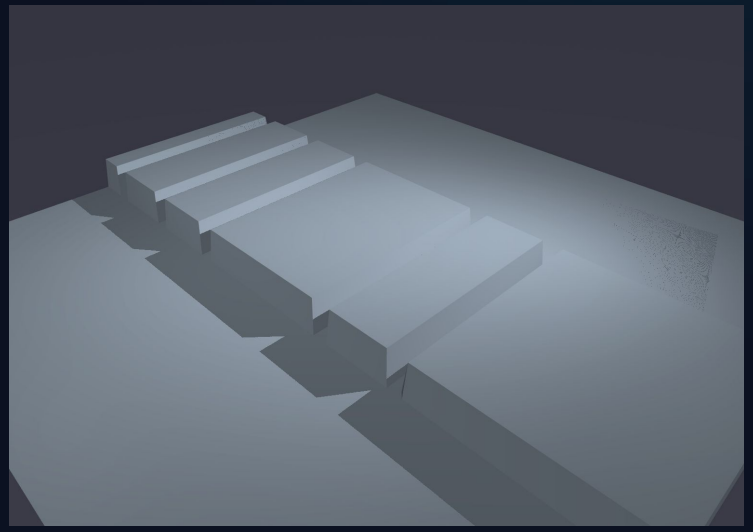
Continuous Hills

Sinusoidal height perturbations regulating terrain unevenness, requiring dynamic center-of-mass adjustments.



Platforms & Hurdles

Elevated discrete flat surfaces arranged as symmetric pyramid stairs testing vertical climbing capability.



Gaps & Trenches

Variable width/depth trenches demanding reach kinematics and coordinated multi-step recovery.

Reward Architecture

🚫 Dense Shaping (Baseline)

Traditional locomotion heavily relies on dense, handcrafted shaping to induce natural gaits.

- Penalizes deviations from nominal base posture (roll, pitch, z-height).
- Strict energy and torque limits (L2 penalties).
- Enforces foot clearance and symmetric phase timing.

Result: High rigidity and vulnerability to specification gaming.

✓ Minimal Formulation (Ours)

Focuses purely on task progression and geometric containment to stay on track.

- **Primary:** Forward velocity task progress bonus.
- **Geometry:** Soft penalties for severe lateral drift to prevent walking off the 4m track.
- **Freedom:** Explicitly excludes pitch penalties, allowing the agent to rear up when climbing hurdles.

Result: Enables emergent leap and recovery behaviors.

Curriculum Algorithms

Domain Randomization

The Baseline

Samples uniformly from the parameter space at each episode. It provides no adaptive curriculum signal, allocating equal training budget regardless of level informativeness or agent capability.

PLR

Prioritized Level Replay

Maintains a replay buffer of levels scored by PVL. Curates difficulty by probabilistically replaying high-regret levels.

ACCEL

Evolutionary Curation

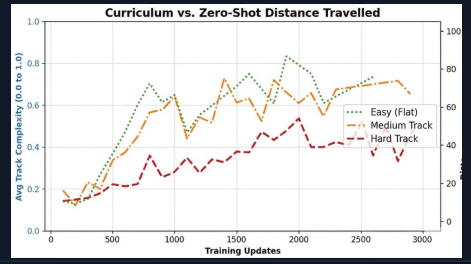
Extends PLR by applying evolutionary edits (Gaussian mutations) to high-regret levels.

Curriculum Algorithms - Training Curves

Domain Randomization

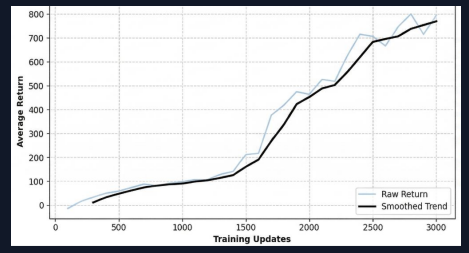
The Baseline





PLR

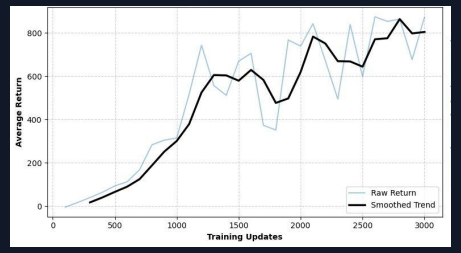
Prioritized Level Replay





ACCEL

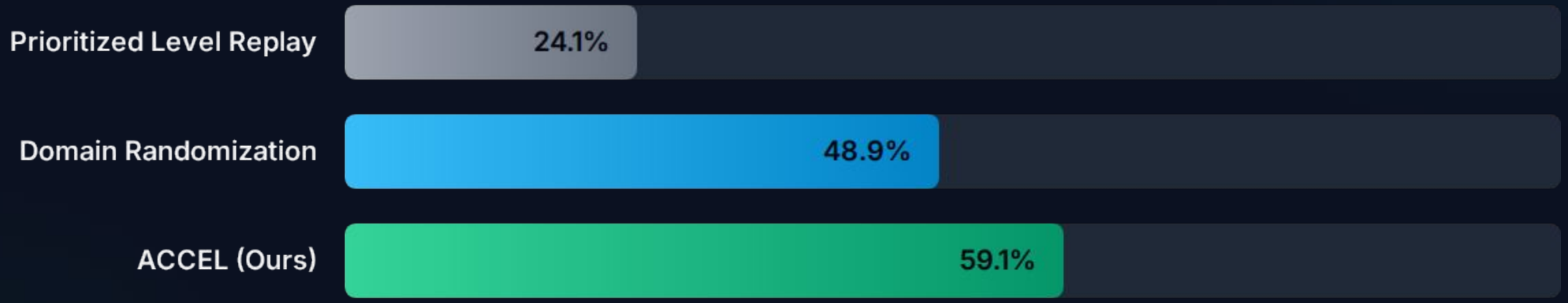
Evolutionary Curation





Suite I: Terrain Generalization (Hard Tracks)

Average solve rate (Normalized Distance %) evaluated on 100 held-out procedurally generated high-complexity tracks.



ACCEL autonomously discovers and amplifies hard configurations through mutation, organically constructing dense obstacle sequences required to train competence on Tier 3 geometries.

Suite II: Dynamic Robustness

Testing zero-shot physical perturbation tolerance on flat terrain to expose the brittleness of dense shaping.

Ice Test (μ=0.1)

Substrate sliding friction drastically reduced from $\mu \approx 1.0$ to $\mu = 0.1$, testing the agent's ability to dynamically widen its stance to prevent lateral slip

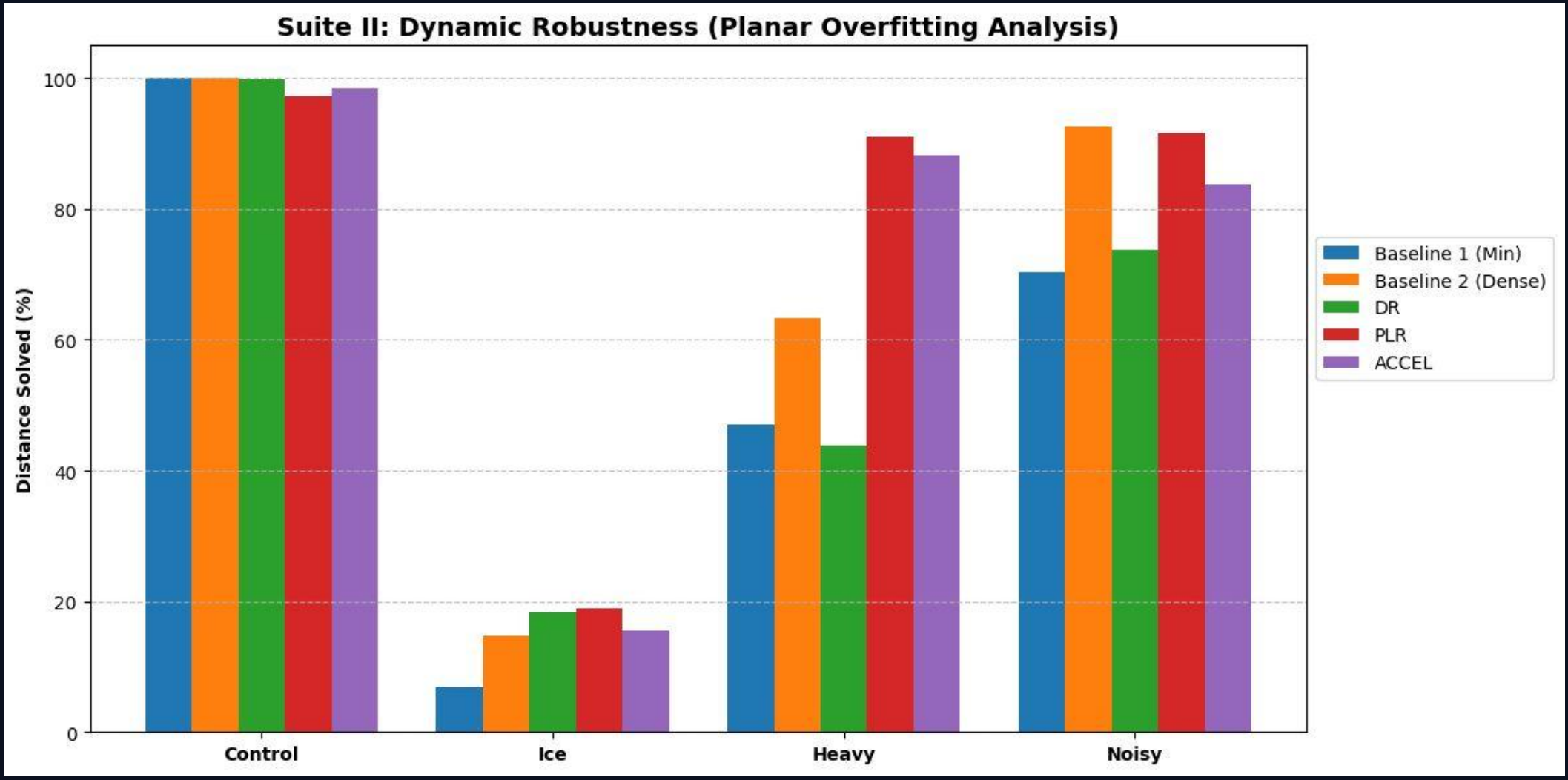
Payload Test (10.0kg)

An unobserved 10.0 kg mass is rigidly attached to the robot's torso, testing torque adaptation and center-of-mass stabilization.

Actuator Noise (25%)

25% random Gaussian noise is continuously injected into the network's action outputs before application to the physics engine, evaluating the policy's real-time error-correction capabilities under severe actuation degradation.

Suite II: Dynamic Robustness



Qualitative Gait Analysis

The Engineered Trot

Baseline 2 (Dense Shaping)

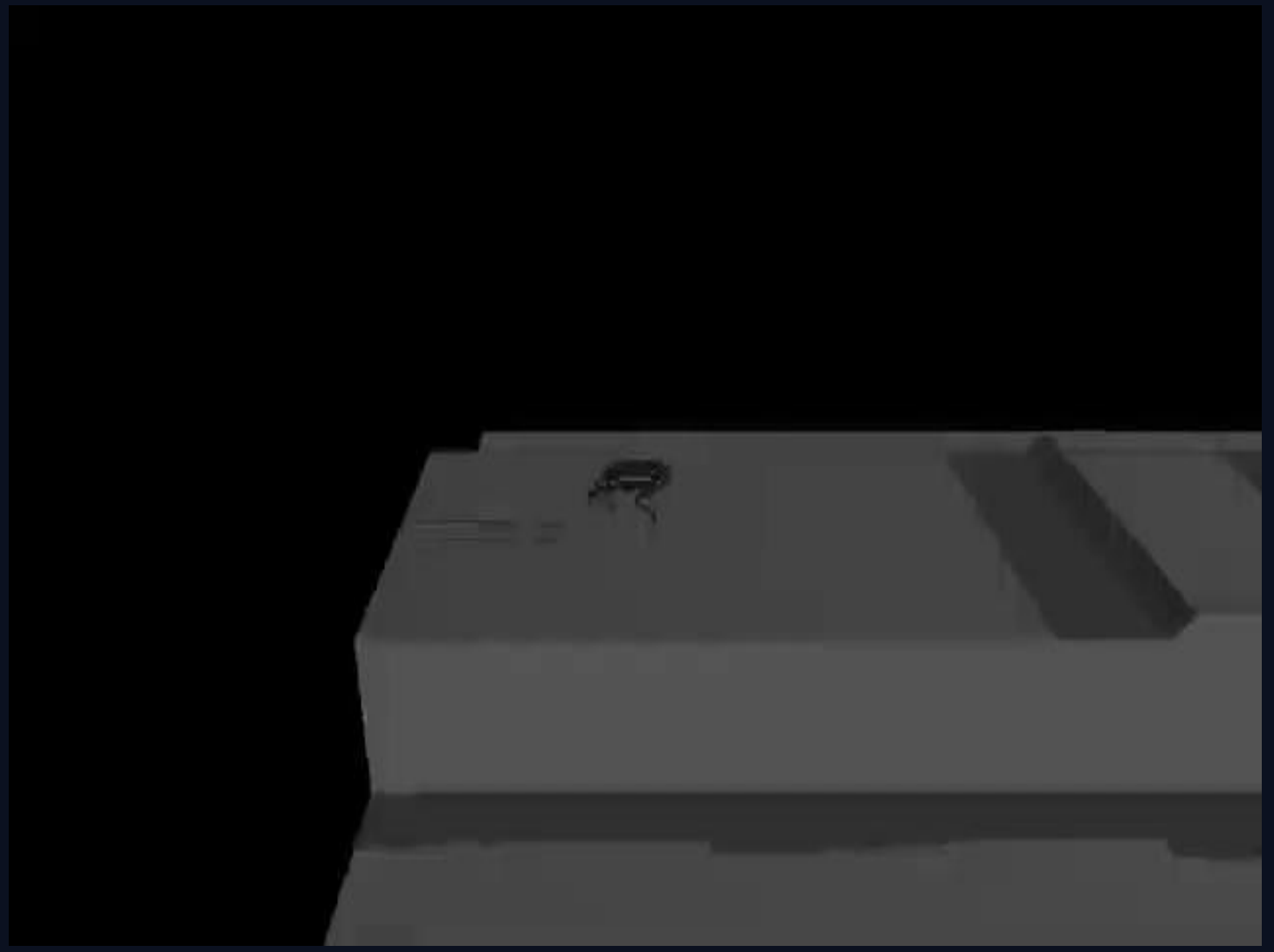
- ✗ Achieves an aesthetically perfect, symmetric trot on flat terrain.
- ✗ Extreme rigidity prevents survival maneuvers; any geometric obstacle causes failure.

Emergent Recovery

ACCEL (Minimal Reward)

- **Approach:** Dynamically lowers center of mass to prepare for jumps.
- **Apex:** Executes synchronized hind-leg propulsion to clear 0.25m hurdles.
- **Recovery:** Utilizes asymmetric stance widening upon landing to absorb impact and maintain momentum.

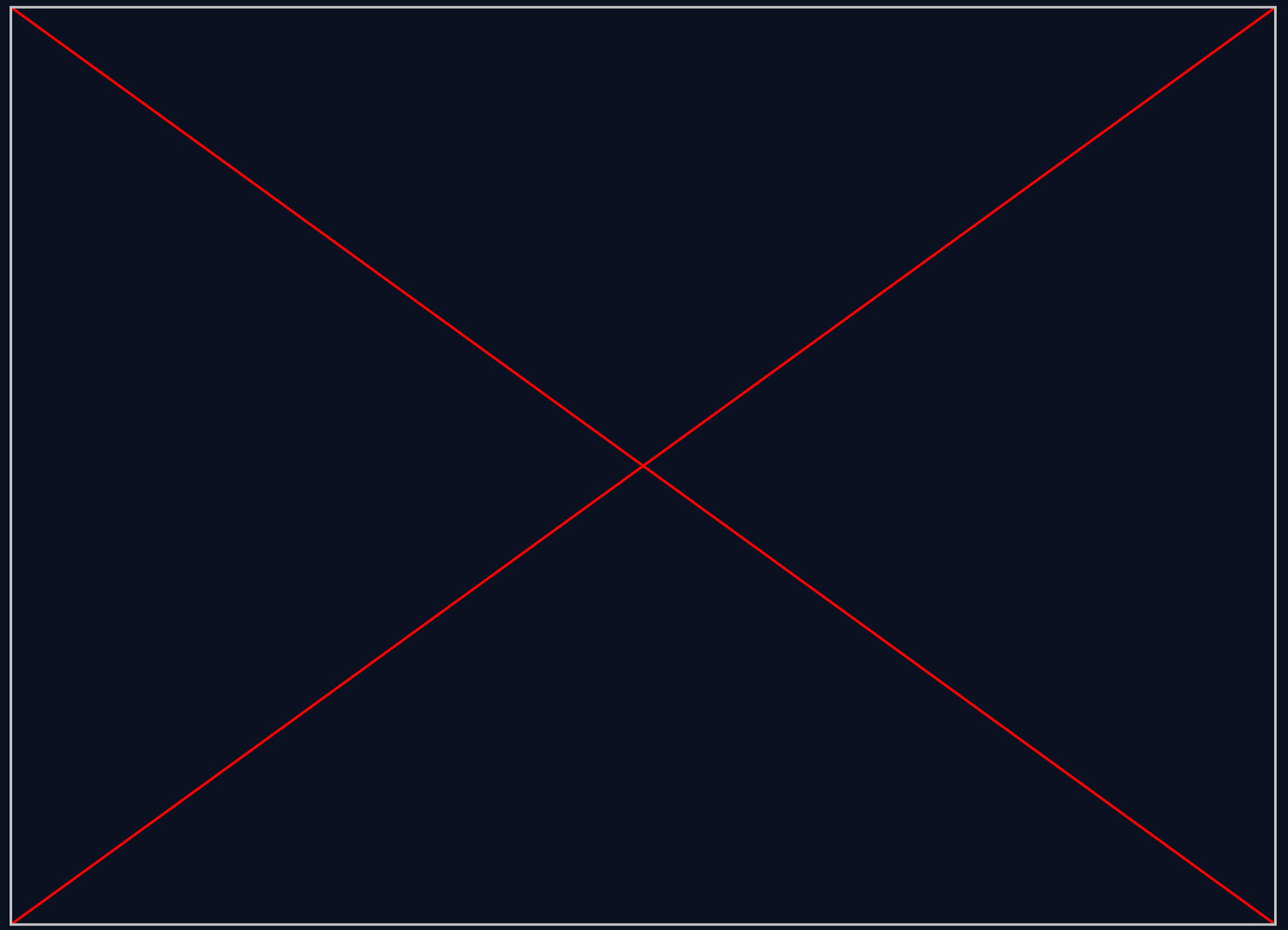
Domain Randomization



Prioritized Level Replay



ACCEL



> Environmental pressure via UED acts as a viable substitute to exhaustive, state-based reward engineering.

Regret-Driven Mastery: Automating curriculum generation through ACCEL continuously scales terrain difficulty, forcing the emergence of highly adaptive recovery behaviors.

Less Shaping, More Robustness: Minimizing dense kinematic rewards in favor of Unsupervised Environment Design improves zero-shot generalization to novel environments.

ADDITIONAL DETAILS

Image Sources



https://www.docs.quadruped.de/projects/go1/html/_images/go1_om_ortho.png

Source: www.docs.quadruped.de



https://thegradient.pub/content/images/2024/11/DALL-E-2024-11-15-15.40.52---Create-an-abstract-image-that-illustrates-how-ReLU-based-neural-networks-shatter-input-space-into-numerous-polygonal-regions--each-behaving-like-a-lin.webp

Source: thegradient.pub



https://www.mdpi.com/sensors/sensors-19-01547/article_deploy/html/images/sensors-19-01547-g003-550.jpg

Source: www.mdpi.com

Appendix: The Regret Problem

Approximating Learnability

UED frames learning as a minimax regret game. Since computing true regret requires knowing the optimal policy, we approximate it using **Positive Value Loss (PVL)**.

$$\mathrm{PVL}(\tau) = \frac{1}{T} \sum_{t=1}^{T} \max(0, \hat{A}_t)$$

Where Ât is the GAE advantage estimate. Positive advantages highlight states where the agent performed worse than expected, signaling high learning potential for a specific terrain.



Appendix: Policy Architecture

A single, non-hierarchical recurrent Actor-Critic architecture trained end-to-end with PPO.

- **Unified State Input:** Concatenates proprioception and exteroception into a 60-dim privileged vector.
- **Shared Embedding:** Processed by a shared dense layer (256 units, ReLU).
- **Memory Core:** Optimized LSTM cell (256 hidden units) enabling temporal tracking of terrain geometry.
- **Decoupled Heads:** Actor outputs 12-dim joint torques; Critic outputs scalar state-value estimates.

